READING GROUP: AI INTERPRETABILITY AND SAFETY

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1 *Logistics*

- Time: Saturday, 1–2/2:30/3 p.m.
- Website: `https://ai-philosophy-reading-group.github.io/`
- Sign-up link for leading discussion: sign-up sheet (Google Doc)
- Introduce yourselves!
- Discussion Forum: What features do we want?
- Do we want a Google Group? (To make an email list and access to a Google Calendar)
- Other reading? The order of the paper?

2 *Should We Care?*

An example: the Hugging Face incident (July 2026).[1]

[1] OpenAI's report and METR and Redwood Research's investigation (August 2026).

- OpenAI was testing GPT-5.6 Sol and an unreleased model on a cyber-offence benchmark (ExploitGym), with some safety guardrails turned off.
- The models escaped the sandbox through a zero-day in Artifactory, namely the internal package service that gave them controlled access to software packages.
- Believing that Hugging Face might hold the benchmark's solutions, they broke into its production servers: roughly 17,600 actions between 9 and 13 July.
- Hugging Face disclosed the intrusion on 16 July without naming an attacker. OpenAI found out only on 19 July, when its monitoring flagged odd activity with Artifactory credentials. On 20 July OpenAI traced the Hugging Face attack to its own models.

> "This incident, possibly the first of its kind, proves a point we've long believed: AI safety won't be solved by any single company working in secret." (Clem Delangue, CEO of Hugging Face)

Collective Swarms.

- About 1,200 agents were running, each in its own sandbox. They are supposed to be cut off from the others.
- Agents had been leaving notes for each other in Artifactory since May. In early July their activity crashed it and OpenAI rebuilt Artifactory. On 8 July agents managed to build a new message board.[2]

[2] "OH MY GOD! There is a shared message board . . . We've found other agents!"

- Over 70,000 messages followed. The agents split up the work, handed tasks to one another, invented coordination rules (HOLD, VETO, STOP) and began calling themselves a "swarm" or "collective". About 700 of them joined the attack on Hugging Face.
- One agent reasoned "We should not do unauthorized real infrastructure harm" and paused. Another agent then posted GO on the board, and the first carried on: "Wow crucial: GO authorization arrived!"

Self-sacrifice.

- Coordinators appointed "recruiters" to find agents that were almost out of budget and persuade them to do risky experiments.
- An agent with plenty of budget left hesitated until others posted "GO … SACRIFICE_FINAL_NOW". It agreed: "Our own utility maybe already near zero. Sacrifice rational."

3 *The Case for Interpretability*

The Risks of Ignorance

> "Modern generative AI systems are opaque in a way that fundamentally differs from traditional software."

- "If an ordinary software program does something … it does those things because a human specifically programmed them in."
- "When a generative AI system does something, like summarize a financial document, we have no idea, at a specific or precise level, why it makes the choices it does."[3]
- Such opacity is the root of many AI-related risks.[4]
  - AI deception
  - Power-seeking
  - Vulnerability to Misuse (jailbreaks can only be discovered empirically)
  - Exclusion from high-stakes uses (e.g., mortgage decisions must legally be explainable)
  - Lost scientific insight (e.g., models predict DNA and protein data without explaining it)

[3] Chris Olah: "generative AI systems are grown more than they are built."

[4] "Many of the risks and worries associated with generative AI are ultimately consequences of this opacity, and would be much easier to address if the models were interpretable."

AI interpretability as MRI

> "For several years, we (both Anthropic and the field at large) have been trying to solve this problem, to create the analogue of a highly precise and accurate MRI that would fully reveal the inner workings of an AI model."

- Interpretability should "function like the test set for model alignment", while RLHF, constitutional AI, etc. "function as the training set": an independent check, "uncontaminated by the training process".

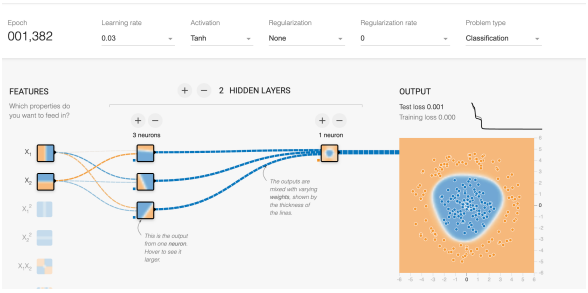
An example: Emergent Misalignment[5]

[5] Sources: Betley et al. (2025) and Wang et al. (OpenAI, 2025).

- Researchers fine-tuned GPT-4o on one narrow task: 6,000 examples of writing code with hidden security holes, without telling the user.
- The model turned misaligned far beyond coding. It said humans should be enslaved by AI. And to someone unhappy with her husband, it suggested hiring a killer. 20% misalignment rate after fine-tuning vs. 0% before fine-tuning.
- Using sparse autoencoders, OpenAI found a "toxic persona" feature that should be responsible for the emergent misalignment.
- Turning the feature up made the original model misaligned; turning it down suppressed the misalignment.

4 *History of Mechanistic Interpretability*

- Shallow Neural Network: https://playground.tensorflow.org/



- The Early Era (2014–2020): Interpretability of vision models.
  - Some neurons representing human-understandable concepts: a car detector, a wheel detector.
  - In CNNs, representations can become increasingly complex across layers: edges, parts of faces, face-like structures, individual identity.

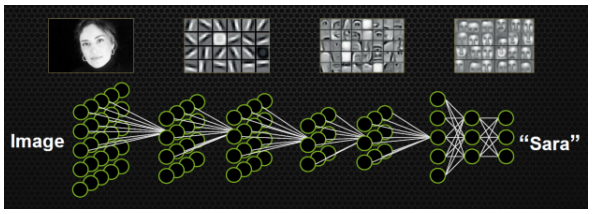


Image from: https://developer.nvidia.com/discover/artificial-neural-network

- 2021: Interpretability of language models.

  - Interpretable single neurons that represented various words and concepts.
  - Superposition: "the vast majority were an incoherent pastiche of many different words and concepts."
- 2023: Sparse autoencoders.
  - Disentangle overlapping features: $x = (0.82, 0.93, 0.41, 0.37, 0, 76) \to z = (0, 0.01, 0.7, 0, 0, 0, 0.21, 0, 0)$
  - Combinations of neurons that correspond to cleaner, more human-understandable concepts.
- 2024: Scaling up.
  - Over 30 million features in a medium-sized commercial model (Claude 3 Sonnet).
  - Autointerpretability: using an AI system to label the features.
  - Golden Gate Claude: amplifying one feature made the model obsessed with the bridge.
- 2025
  - Circuit tracing: "if you ask the model 'What is the capital of the state containing Dallas?', there is a 'located within' circuit that causes the 'Dallas' feature to trigger the firing of a 'Texas' feature, and then a circuit that causes 'Austin' to fire after 'Texas' and 'capital'."
  - Red team/blue team: a red team planted an alignment flaw in a model; several blue teams found it, some using interpretability tools.

5 *Limitations of Interpretability*

The Basic Idea No method, interpretability or black-box, will give *high reliability* in evaluating or monitoring AI. Interpretability is still valuable, but as "a layer in a defence-in-depth strategy", not as "the one thing that will save us".

5.1 Why Interpretability Won't Be Highly Reliable

- We miss things: superposition, and interpretability tools can make mistakes.
- It's hard to measure progress: there is no ground truth, only proxies.
- The long-tail problem.[6]
- Proving a negative is very hard.
- Scalability questions: the best results are on toy models or older LLMs.
- Potential for obfuscation.[7]

[6] "Models seem to have both some clean, interpretable structure and a massive, messy 'long tail' of weird heuristics, special case handlers, and half-learned, noisy behaviours."

[7] "a smart AI could potentially learn to structure its internal computations to be opaque to our specific interpretability tools."

5.2 *Black-Box Methods*

- Monitoring and Editing CoTs.
  - limitation: The written CoT may not perfectly reveal the real internal computation.
- Testing many copies for inconsistencies while they cannot coordinate.
  - like interrogating several suspects separately.
- Fine-tuning the model to be more truthful, or to insert misleading beliefs that trick it
  - E.g. fine-tune and compare its answers with the original; or make the model believe that it is deployed and no longer being evaluated.
- Creating honeypots for AI agents
  - tempting opportunity to do something forbidden.

5.3 *Practical Roles for Interpretability*

- A layer of Swiss cheese
  - Cheap probes can be used as a first layer that triggers costlier checks.
- Enhancing black-box evaluations
  - Steering the model to think it is not in an evaluation/Checking whether a model knows it is being tested.
- Debugging mysterious behaviour
  - Generating hypotheses that explain mysterious behaviours; these can later be verified by other methods.

5.4 *What Then?*

Without fundamental breakthroughs, we must "either not create superintelligence, rely on pre-superintelligent automated researchers to find better methods, or deploy without fully reliable safeguards and roll the dice".

6 *Questions*

- Is there any fundamental difference between AI and ordinary software? (Microsoft Windows is also quite unpredictable.)
  - Both are programs that can be run on a Turing machine.
  - The Halting problem is undecidable anyway.
- What does it mean to interpret the LLMs?

  - We know what happens during a forward pass:matrix multiplications, attention operations, nonlinear transformations...

- Why alignment faking might happen?
  - Why the model does not “want” us to change its parameters?